



Daily rhythms of nonshivering thermogenesis in common spiny mice *Acomys cahirinus* under short and long photoperiods

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Abstract

Daily rhythms of nonshivering thermogenesis NST were studied in common spiny mice *Acomys cahirinus*, acclimated to different photoperiod regimes (16L:8D and 8L:16D) at a constant ambient temperature of 26°C. Noradrenaline NA (1.5 mg/kg subcutaneous) was injected at: 06:00, 12:00, 18:00 and 24:00 h (± 15 min). NST was measured as the ratio between the maximal oxygen consumption (VO_2) as response to NA — VO_2NA and VO_2 measured at 26°C — $\text{VO}_{2\text{min}}$. Rectal temperatures T_{bNA} and T_{bmin} respectively were recorded at the end of VO_2 measurements. Significant variations in T_{bmin} , T_{bNA} , and NST were revealed, under the two different photoperiod regimes. Significant differences in $\text{VO}_{2\text{min}}$, NST, T_{bmin} and T_{bNA} were also recorded within each photoperiod acclimation group. These results suggest that daily and photoperiod depended variations in the brown adipose tissue activity, presumably emerge from amount of unoccupied receptors or changes in the receptors affinity to NA. © 2000 Published by Elsevier Science Ltd. All rights reserved.

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1. Introduction

Nonshivering thermogenesis (NST) is an important mechanism for heat production in cold and short photoperiod acclimated small rodents, as well as in winter acclimatized ones. The main advantage of this mechanism is in the generation of a great amount of heat over a short period without any interference with motor activity. This mechanism can be activated rapidly (minutes) as it is under the control of the sym-

pathetic nervous system (Jansky, 1973). This mechanism is also important in desert rodent species, which have low metabolic rates, lower than the expected for their body mass, although they are not cold or short photoperiod acclimated (Haim and Izhaki, 1993). In a study carried out on golden spiny mice *Acomys russatus*, a NST daily rhythm measured as a maximal response to a noradrenaline (NA) injection was noted. Furthermore, a minimal oxygen consumption ($\text{VO}_{2\text{min}}$) daily rhythm was also recorded. The maximal $\text{VO}_{2\text{min}}$ values were at 06:00 and 18:00 h while the maximal VO_2NA values were recorded in the dark phase at 24:00 h (Kronfeld et al., 1994). In a study carried out on the wood mouse *Apodemus sylvaticus*, a NST daily rhythm was noted but it was also revealed that this

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rhythm responded to changes in photoperiod (Haim et al., 1995).

The common spiny mouse *A. cahirinus* and the golden spiny mouse *A. russatus* coexist in the extreme hot and arid parts of the Rift Valley in Israel. Their coexistence is through temporal segregation where *A. cahirinus* excludes *A. russatus* from nocturnal activity. However, in such mice, although they show a body temperature (T_b) and activity daily rhythms, maximal T_b values are recorded during the dark phase (Haim and Zisapel, 1995). The T_b daily rhythm of long photoperiod (16L:8D) acclimated *A. cahirinus* was studied (Haim and Zisapel, 1994). From the results of this study it was noted that minimal T_b values were recorded during the light phase (between 12:00–18:00 h) while maximal T_b values were obtained during the dark phase (between 24:00–06:00 h).

The aims of the present study were (1) to inquire if common spiny mice have a NST daily rhythm, (2) to see if this rhythm responds to photoperiod manipulations.

2. Materials and methods

2.1. Animals

Common spiny mice ($n = 24$, males and females) were obtained from a breeding colony, kept at the Department of Biology, University of Haifa, Oranim. The tested mice were progenies of parents collected at Ein Gedi close to the Dead Sea shores. Each mouse was kept separately in a cage with sawdust used as bedding. All mice were acclimated for at least 3 weeks to long (LD) and short (SD) photoperiod (16L:8D LD and 8L:16D respectively, starting with LD-acclimation) at a constant ambient temperature (T_a) of 26°C. A bright light (440 lx) was used during the light phase, and a dim red light (20 lx) was kept on during the light and dark period, in order to facilitate handling of the mice during the dark period. The mice were fed ad libitum on a pelleted rat diet with carrots added as a source of moisture. Food was available until the beginning of NST measurements.

2.2. Nonshivering thermogenesis (NST)

Measurements were carried out inside a metabolic chamber (volume 800 ml), using an open flow system (Depocas and Hart, 1957; Hill, 1972), with a flow of dried air (Silica gel, TamRod) at a rate of 600 ml/min. The metabolic chamber was submerged into a regulated water bath (Neslab 500) with a connected Exocal system for cooling the water. Oxygen concentration of the dried excurrent air was determined by using an Applied Electrochemistry A3 oxygen analyzer and

monitored on a Tek-Dyn-712 recorder as in Haim (1987).

At the first stage oxygen consumption VO_2 was measured for 30 min after a stabilizing period of about 2 h — $VO_{2\min}$. The mouse was then removed from the metabolic chamber, T_b was measured $T_{b\min}$ and nor-adrenaline (NA) (1.5 mg/Kg, Sigma) was injected (s.c.) at intervals of 6 h (06:00, 12:00, 18:00 and 24:00, ± 15 min). Each individual was injected once only. At the second stage the mouse was returned to the metabolic chamber and VO_2 measurements went on until 10 min after values reached a peak (VO_{2NA}). Then the mouse was removed from the metabolic chamber and T_{bNA} was recorded.

Body (rectal) temperatures were measured by using a copper-constantan thermocouple, and monitored on a TH-65 Wescor thermometer. It was measured by quickly (30 s) inserting the thermocouple 3 cm into the rectum of the mouse.

NST was calculated from the ratio between VO_{2NA} and $VO_{2\min}$, measured at $T_a = 26^\circ\text{C}$, in unanaesthetized mice, as in (Haim and Martinz, 1992).

2.3. Statistics

All values are given as mean $\pm$ SD. The data within each group of photoperiod acclimation and between two groups of acclimation were analyzed by using 2-way analysis of variance (ANOVA). In the case of $VO_{2\min}$ and VO_{2NA} the data within each group of photoperiod acclimation was analyzed by 1-way ANOVA. The ANOVA was followed by a posteriori Scheffer pairwise comparison of means ($P < 0.05$).

3. Results

The effects of NA injections at different hour of the day on VO_2 in LD and SD acclimated *A. cahirinus* are shown in Fig. 1. The 2-way ANOVA did not explore significant differences in VO_2 using the combination of different hours and photoperiod response. However, a 1-way ANOVA test revealed a significant difference in $VO_{2\min}$ between hours, within each group. LD-mice had significantly ($F_{3,20} = 18.446$; $P < 0.001$) the lowest $VO_{2\min}$ at 18:00 h, while SD-mice had significantly ($F_{3,20} = 12.601$; $P < 0.01$) the highest $VO_{2\min}$ values at the same time. VO_{2NA} values in LD-mice were low at 12:00 h while for SD-mice low values were recorded at 24:00 h but in both cases, they did not differ significantly from the other hours.

Analyzing the $T_{b\min}$ by 2-way ANOVA revealed that time of measuring within each group, is significant ($F_{1,44} = 5.443$; $P < 0.05$) under both photoperiod treatments. While in LD-mice the lowest $T_{b\min}$ values were recorded at 18:00 h in SD-mice they were recorded at

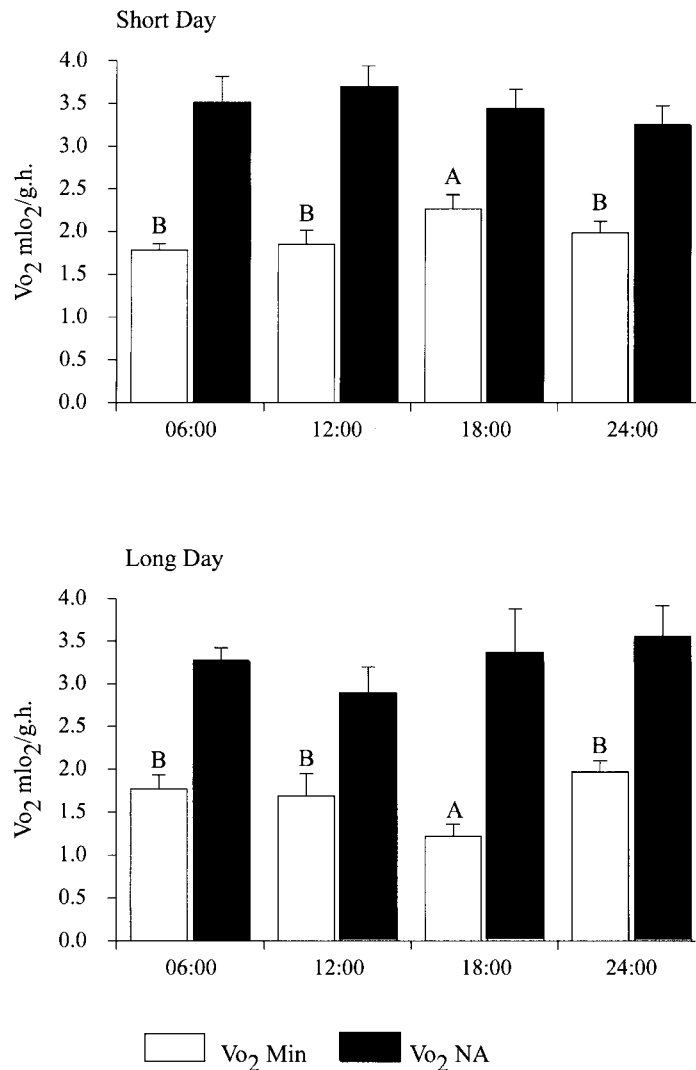


Fig. 1. Daily changes in oxygen consumption (VO_2) measured as resting metabolic rates ($VO_{2\min}$) and maximal response to NA VO_{2NA} , in *Acomys cahirinus*, under two photoperiod regimes. $VO_{2\min}$ was measured at an ambient temperature of 26°C while VO_{2NA} was measured after a NA (1.5 mg/kg, s.c) injection. All values are mean $\pm$ S.D. for $n = 6$ individuals. Under short day acclimation (SD) $A > B$ while under long day (LD) $A < B$. $VO_{2\min}$ values measured at 18:00 h SD-mice, are significantly ($P < 0.01$) higher compared to LD-mice.

12:00 h (Table 1). This is not the case for T_b NA in which the 2-way ANOVA revealed that photoperiod acclimation (treatment) and the interaction with time were significant ($F_{1,44} = 11.172$ and 12.649 ; $P < 0.005$ and 0.001 respectively). However, using a 1-way ANOVA revealed within LD-mice that T_bNA was significantly ($F_{3,20} = 21.6$; $P < 0.001$) lower at 12:00 h (Table 1).

As in 2-way ANOVA for NST values only the interaction between hours within each group and treatment (photoperiod acclimation) was significant ($F_{1,44} = 5.557$; $P < 0.05$), data for each treatment was

analyzed by 1-way ANOVA. The results revealed that the highest NST values ($F_{3,20} = 22.9$; $P < 0.001$), within LD-mice were at 18:00 h, while in SD-mice highest values ($F_{3,20} = 21.6$; $P < 0.001$) were measured at 06:00 and 12:00 h (Table 1).

4. Discussion

The results of the present study show diurnal variations of NST in *A. cahirinus* which manifest a daily rhythm. Small rodents as do other small mammals, use

Table 1

Daily changes in body temperature (T_b). $T_{b\min}$ measured at the end of resting metabolic rates ($VO_{2\min}$) measurements, T_{bNA} measured at the end of VO_{2NA} measurements, the maximal response to NA injection and nonshivering thermogenesis (NST) the ratio between VO_{2NA} and $VO_{2\min}$. All values are mean $\pm$ S.D. for $n = 6$ individuals. In short day (SD) acclimated mice $A > AB > B$ while for long day acclimated mice for T_b $B < A < BC < AB < C$ while for NST in this group $B > A$

Hour	$T_{b\min}$ ($^{\circ}\text{C}$)	T_{bNA} ($^{\circ}\text{C}$)	NST
<i>Short day</i>			
06:00	37.0 ± 0.3^B	38.0 ± 0.3^{AB}	1.97 ± 0.1^A
12:00	36.8 ± 0.4^B	38.2 ± 0.3^A	2.00 ± 0.1^A
18:00	38.3 ± 0.4^A	37.5 ± 0.3^B	1.52 ± 0.1^B
24:00	37.8 ± 0.4^A	37.3 ± 0.3^B	1.64 ± 0.1^B
<i>Long day</i>			
06:00	37.2 ± 0.5^A	38.0 ± 0.4^{AB}	1.85 ± 0.2^A
12:00	36.4 ± 0.4^B	37.2 ± 0.5^A	1.72 ± 0.2^A
18:00	35.8 ± 0.4^B	37.5 ± 0.3^{BC}	2.75 ± 0.3^B
24:00	38.0 ± 0.3^A	38.6 ± 0.3^C	1.80 ± 0.2^A

NST to increase their heat production upon exposure to low ambient temperatures. Such an NST is defined as regulatory NST (Jansky, 1973). NST values were found to be higher in cold acclimated rodents (Heldmaier, 1972; Jansky, 1973) as well as in short day acclimated rodents although they were kept in relatively high ambient temperatures (Lynch, 1970; Haim et al., 1980; Heldmaier et al., 1981; Haim, 1982; Ellison et al., 1992).

Daily rhythms of body temperature have been observed so far in many rodent species (Heldmaier et al., 1989; Refinetti and Menaker, 1992). Heat production daily rhythms in rodents were found to depend on relation to their body mass (Aschoff, 1982) and body temperature (Haim et al., 1988; Rubal et al., 1992; Haim and Zisapel, 1995). As heat production and body temperature change on the basis of a daily rhythm, it is not surprising that NST also changes on such a basis. Indeed, in golden spiny mice *A. russatus* (Kronfeld et al., 1994) and in wood mice *A. sylvaticus* (Haim et al., 1995), NST daily rhythms were observed. Recently, Gebozynski and Taylor (in press) reported that they were not able to demonstrate regulatory NST daily rhythms in the laboratory mouse *Mus musculus*. However, significantly daily changes in regulatory NST, measured as a response to NA (0.66 mg/kg) injection were demonstrated in the root vole *Microtus oeconomus*. The NST rhythm in the vole occurred, despite the fact that the voles did not exhibited a clear T_b daily rhythm while the mice did.

In our study the same dosage of NA (1.5 mg/kg) was injected to the LD-mice and SD-mice, at the different hours of the day. As $VO_{2\min}$ values differed

significantly between groups and within each group while VO_{2NA} values were similar (Fig. 1) the difference in NST values (Table 1) emerge from different rates of VO_2 increase. Therefore, it would be reasonable to assume that the sensitivity of the adrenergic receptors, on brown adipose tissue (BAT), in mice with low $VO_{2\min}$ values were higher than those with relatively higher $VO_{2\min}$ values. An alternative explanation could be related to differences in receptor numbers occupied by endogenous NA, between mice with relatively low or high $VO_{2\min}$ values.

The daily changes of brown adipose tissue (BAT) thermogenesis in newborn rats were studied by Redlin et al. (1992). The thermogenic state of BAT of such rats was tested during minimal and maximal phases of the T_b cycle using the binding of guanosine 5'-diphosphate (GDP) to mitochondrial BAT as a thermogenic activity index. A significant decrease in GPD binding activity accompanied T_b decrease, during the resting phase in the daily cycle.

Our results further support the idea that daily variations in BAT activity on the one hand and the response to photoperiod manipulations on the other, are presumably through changes in the receptors affinity to NA. These changes in affinity may be the cause, for the daily variation of the response to NA injection. Furthermore, these variations are presumably manifested as a daily NST rhythm, in common spiny mice. However, a difference in the number of occupied receptors by endogenous NA should not be ruled out.

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